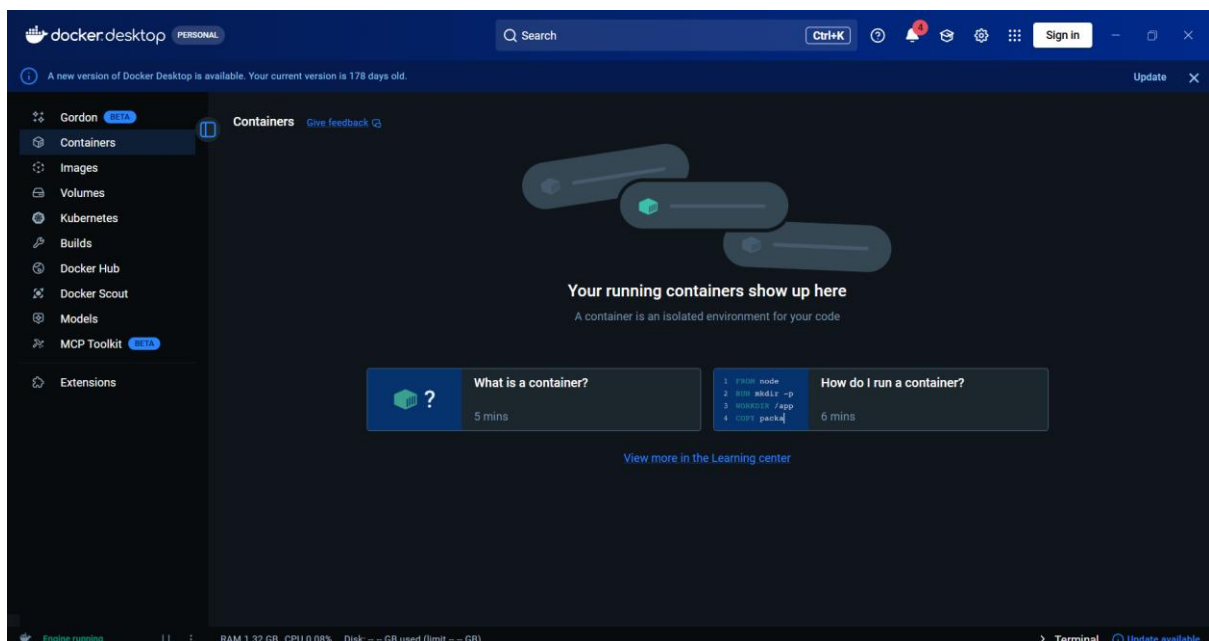


Name: P. Kavya Sai
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Section: 9

ASE TASK-6

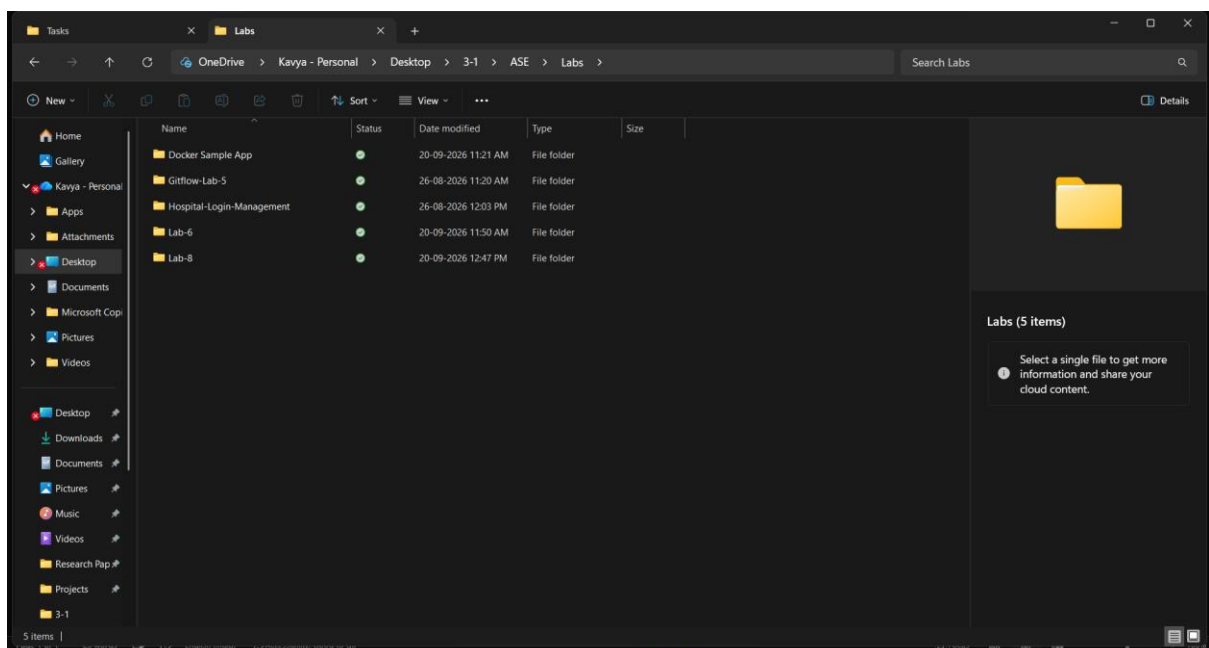
DOCKER CONTAINERIZATION

1) Download and open the docker desktop and wait until it status is running

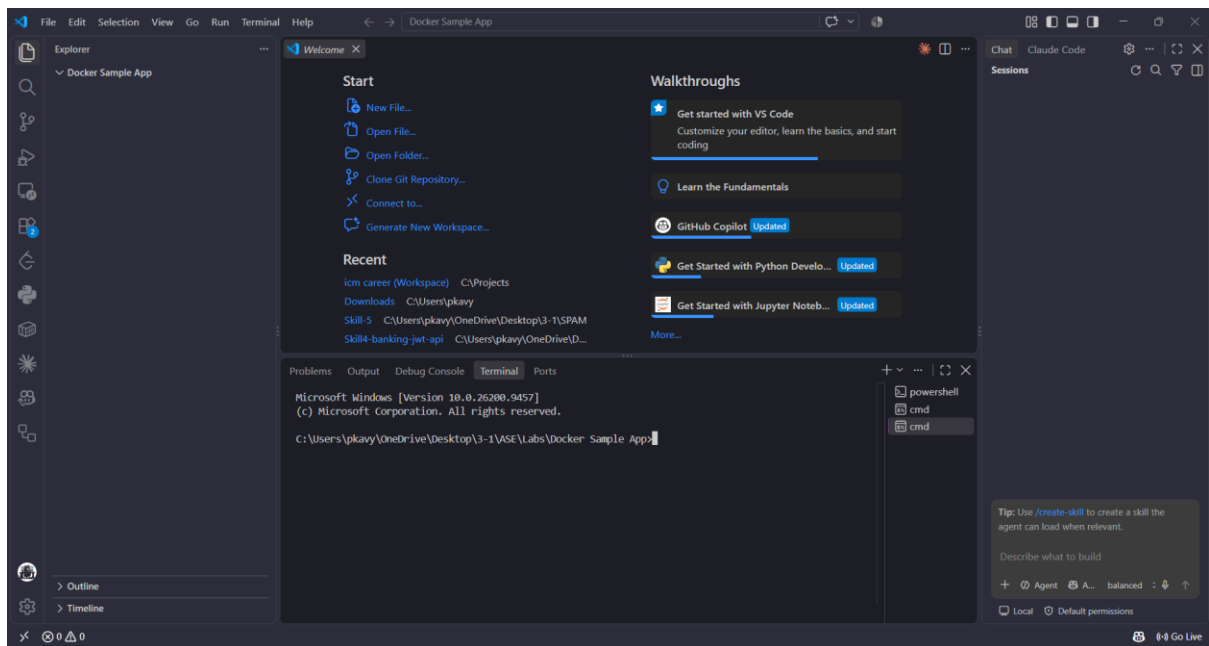


2) Create a project structure

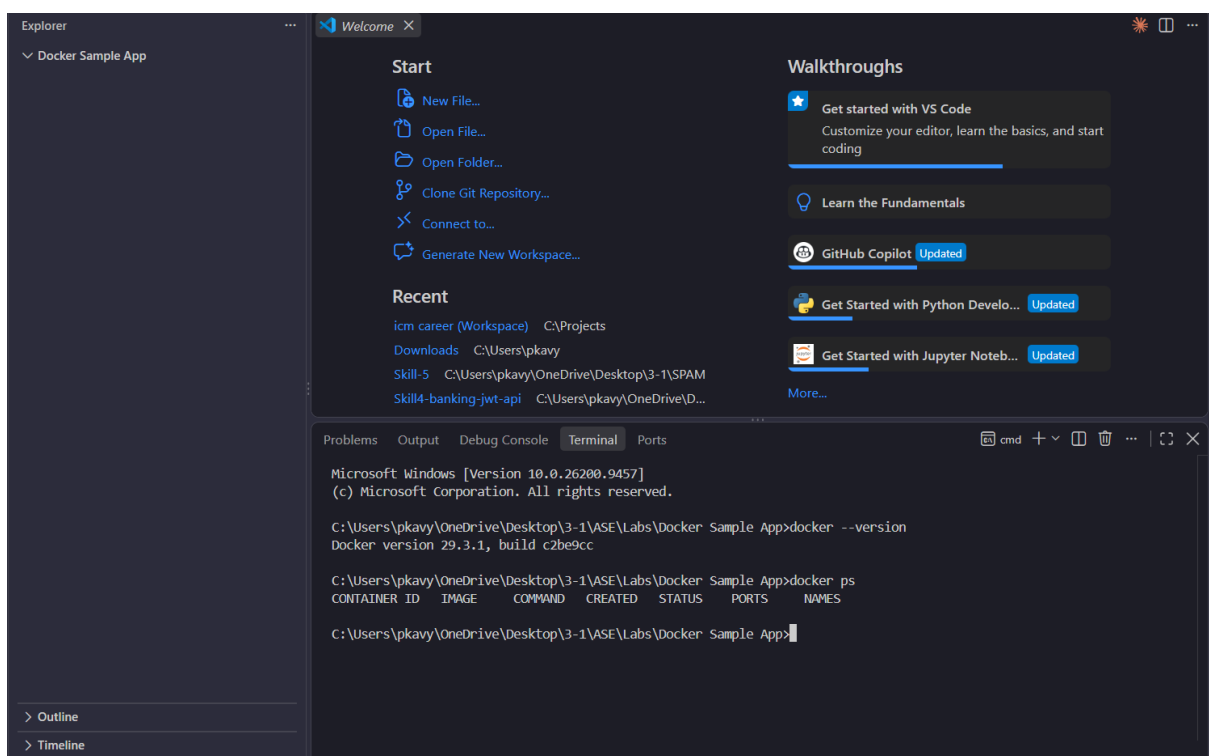
->Create a folder named docker-sample-app



➔ Open the folder in the vs code



➔ Check if the docker is running in your terminal

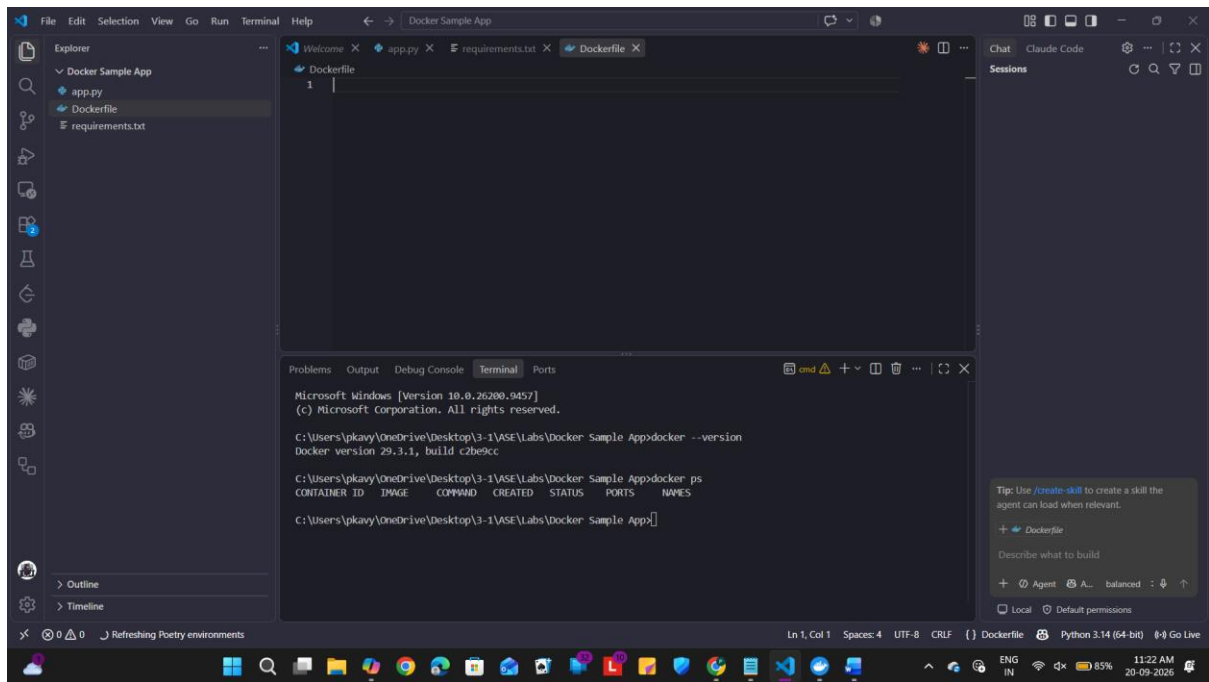


➔ Create the following files:

app.py

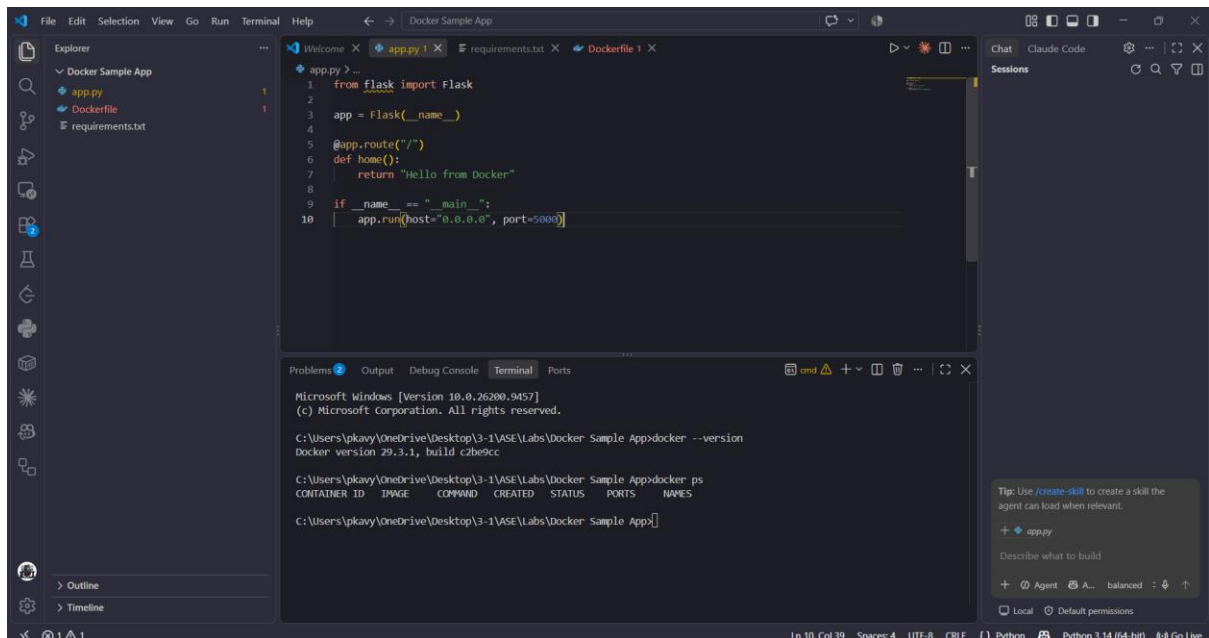
Dockerfile

requirements.txt



3) Create a flask application

- ➔ Open the app.py file
- ➔ import flask
- ➔ Create a flask application
- ➔ Define the route
- ➔ Make the route return “Hello from Docker!”

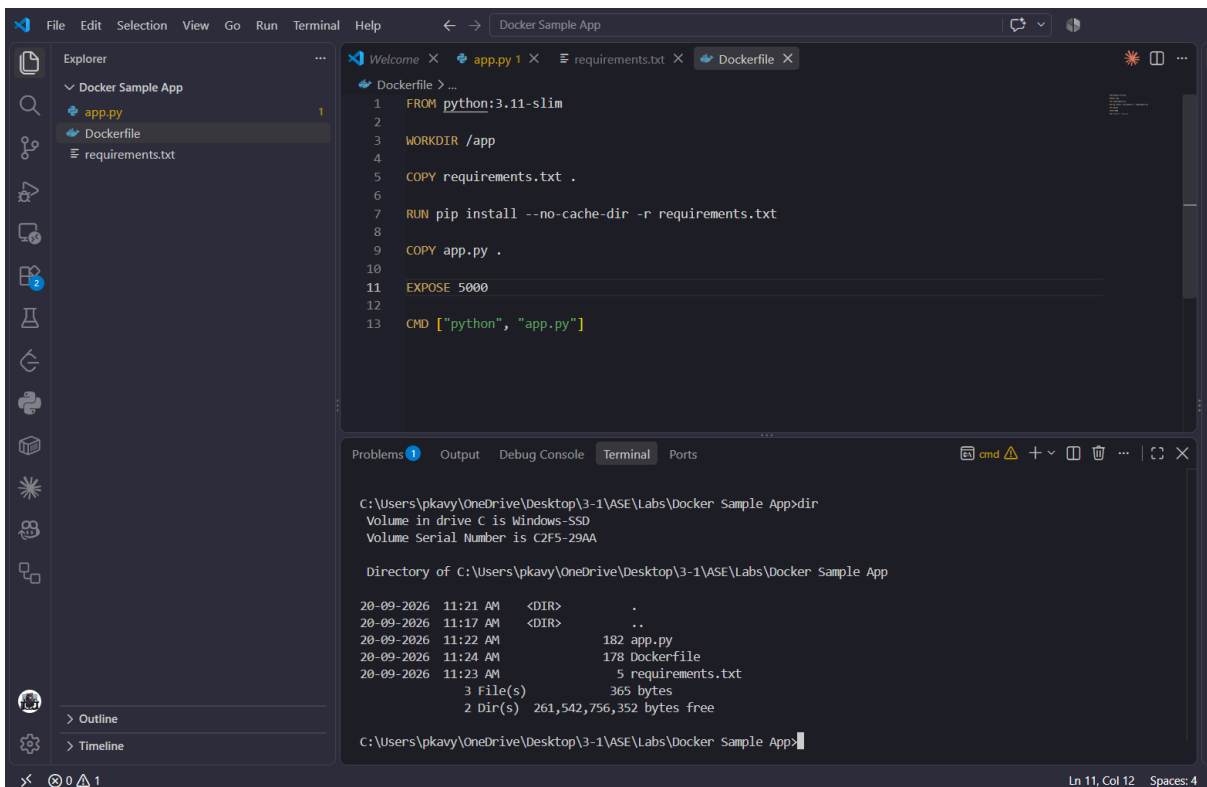
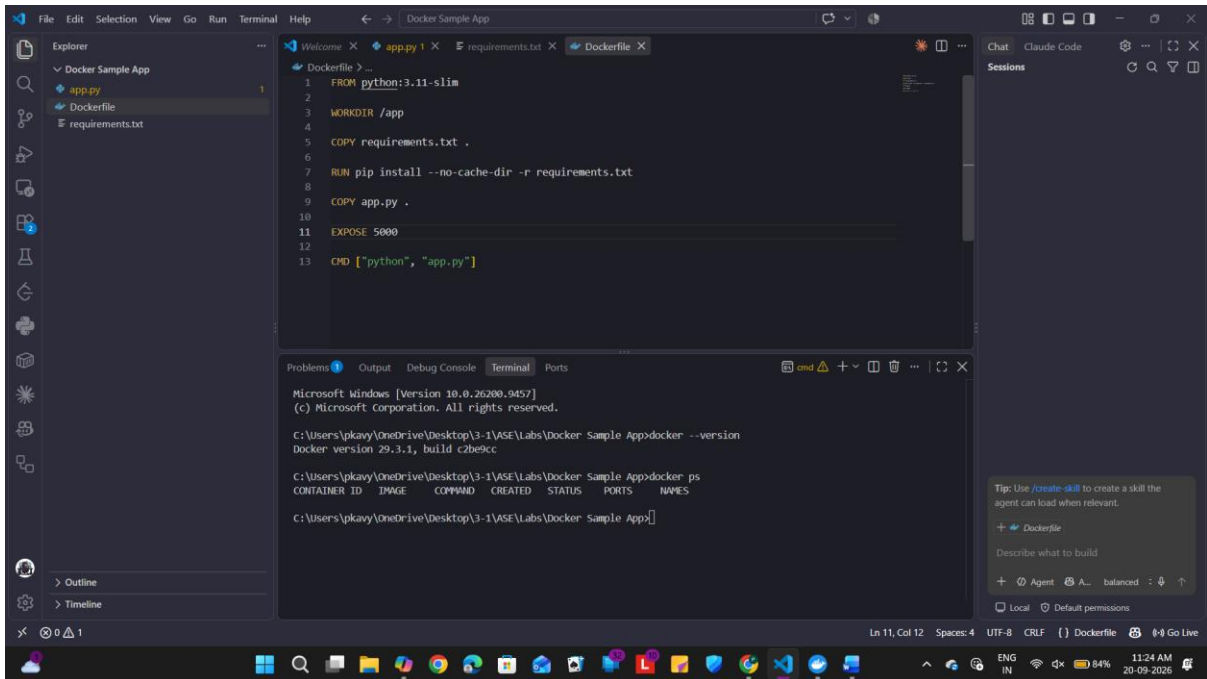


3) Create the dockerfile

- ➔ Create a file dockerfile without any extension
- ➔ Specify the python base image

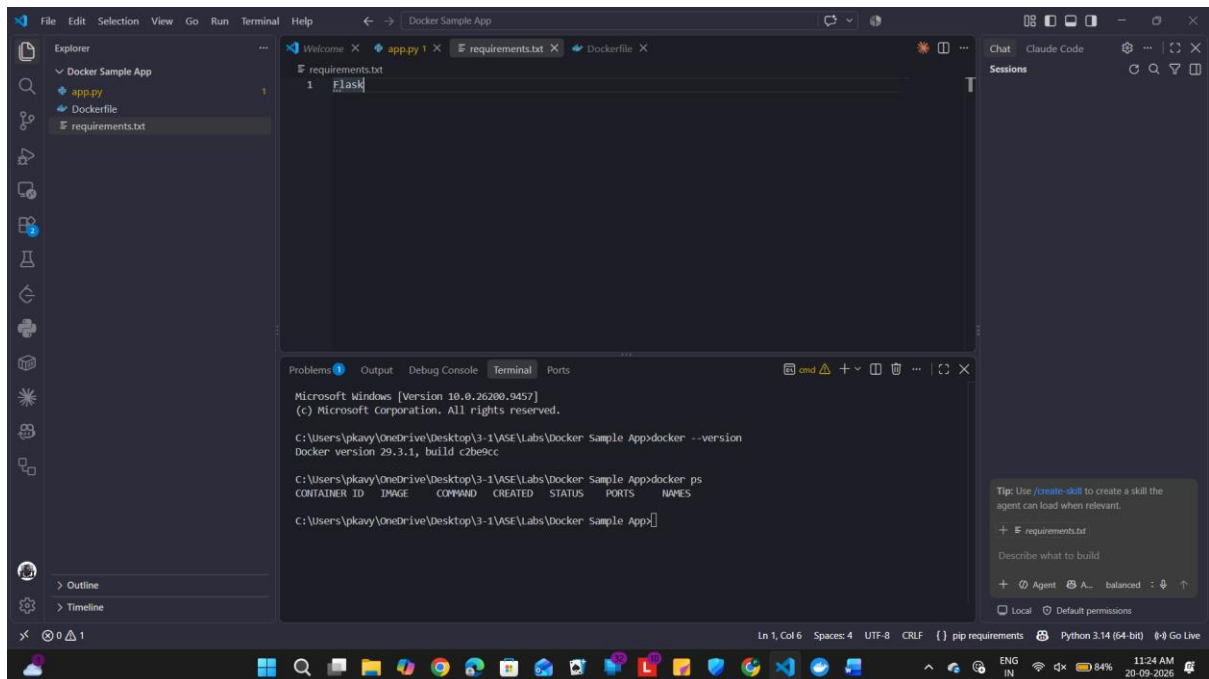
- ➔ Set the working directory to /app
- ➔ Specify the command to start the flask application

```
C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>docker build -t my-flask-app:1.0 .
```



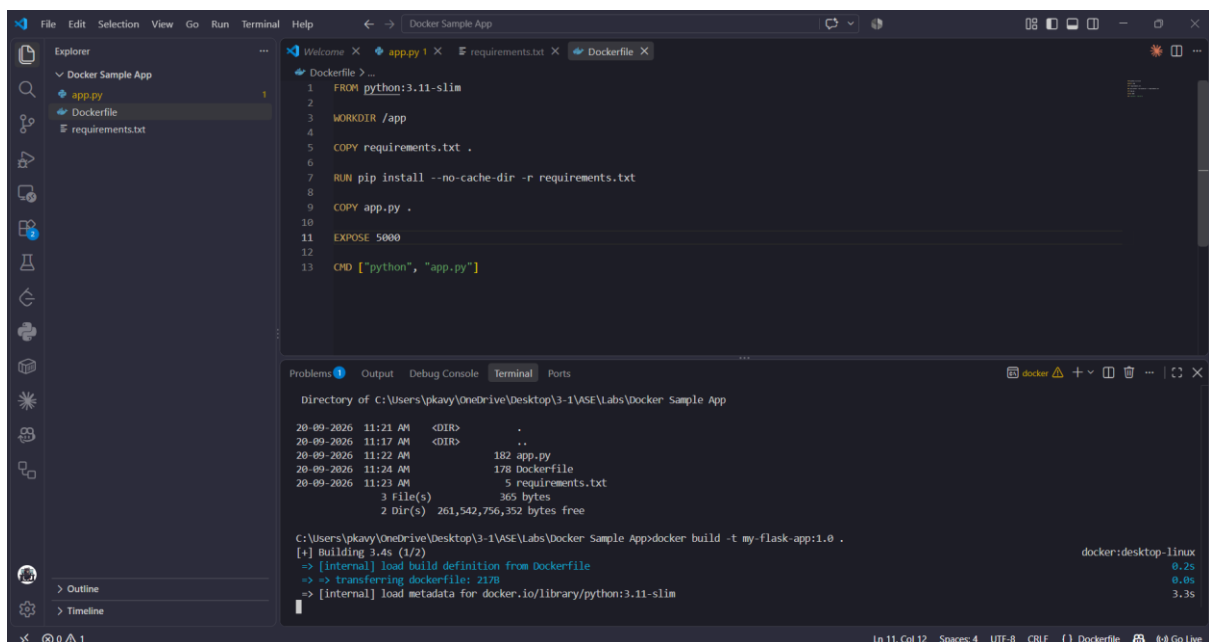
4) Create requirements.txt

- ➔ Open requirements.txt. Add the required python package. Save the file and build the docker image.



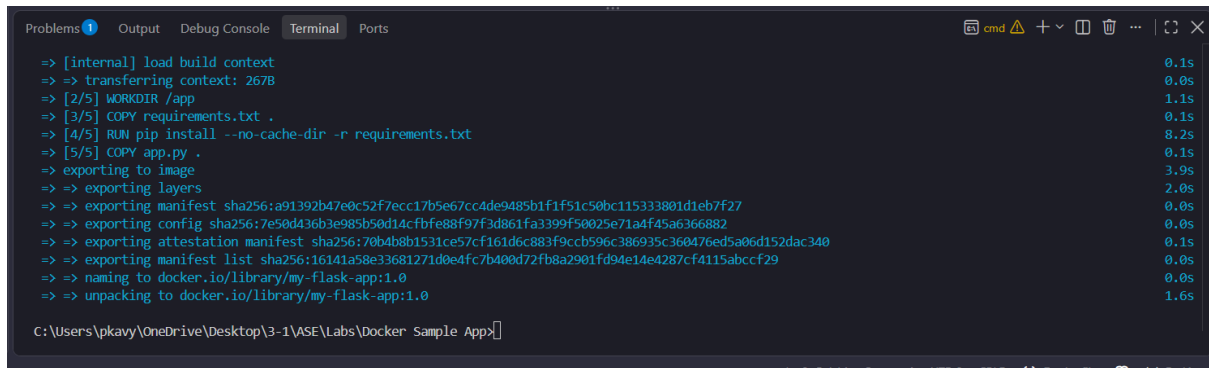
5) Verify the docker image

- ➔ Open the terminal in the required directory
- ➔ Execute the command `docker build -t my-flask -app: 1.0`



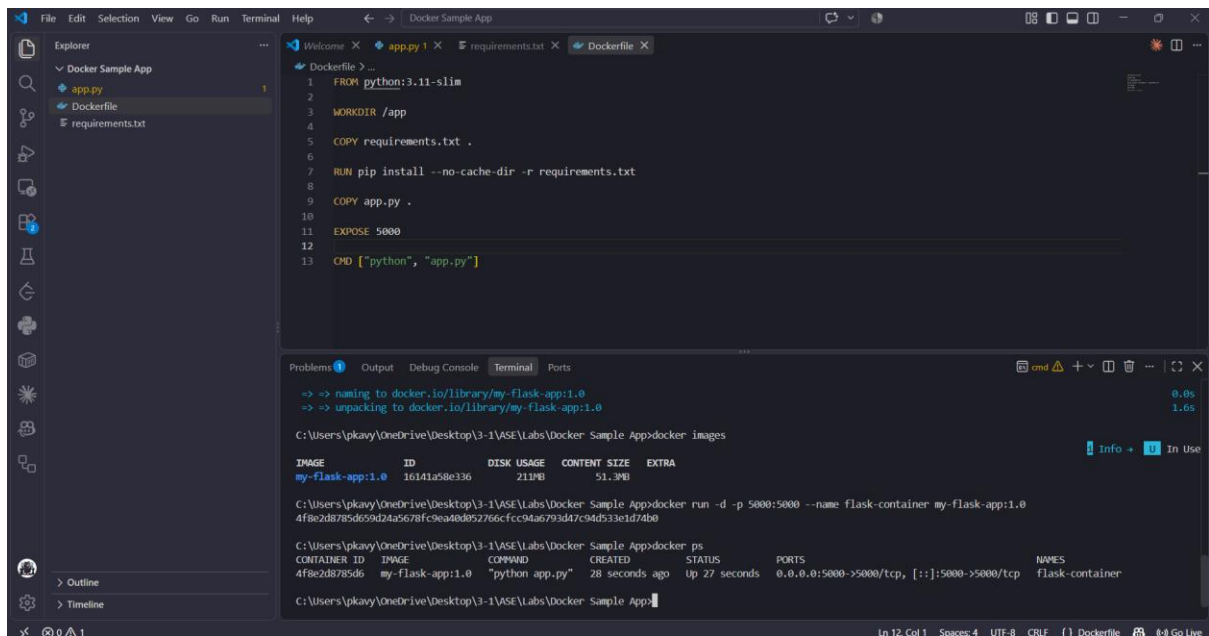
- ➔ Execute docker image

- ➔ ->Verify that
my-flask-app with tag 1 0 is present



```
-> [internal] load build context
-> => transferring context: 267B
-> [2/5] WORKDIR /app
-> [3/5] COPY requirements.txt .
-> [4/5] RUN pip install --no-cache-dir -r requirements.txt
-> [5/5] COPY app.py .
-> exporting to image
-> => exporting layers
-> => exporting manifest sha256:a91392b47e0c52f7ecc17b5e67cc4de9485b1f1f51c50bc115333801dieb7f27
-> => exporting config sha256:7e50d436b3e985b50d14cfbf88f97f3d861fa3399f50025e71a4f45a6366882
-> => exporting attestation manifest sha256:70b4b8b1531ce57cf161d6c883f9ccb596c386935c360476ed5a06d152dac340
-> => exporting manifest list sha256:16141a58e33681271d0e4fc7b400d72fb8a2901fd94e14e4287cf4115abccf29
-> => naming to docker.io/library/my-flask-app:1.0
-> => unpacking to docker.io/library/my-flask-app:1.0

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>
```



```
FROM python:3.11-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py .
EXPOSE 5000
CMD ["python", "app.py"]

-> => naming to docker.io/library/my-flask-app:1.0
-> => unpacking to docker.io/library/my-flask-app:1.0

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>docker images

IMAGE          ID              DISK USAGE    CONTENT SIZE  EXTRA
my-flask-app:1.0 16141a58e336    211MB         51.3MB

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>docker run -d -p 5000:5000 --name flask-container my-flask-app:1.0
4f8e2d8785d659d24a5678f9c9a40d052766cfc94a6793d47c94d533e1d74b0

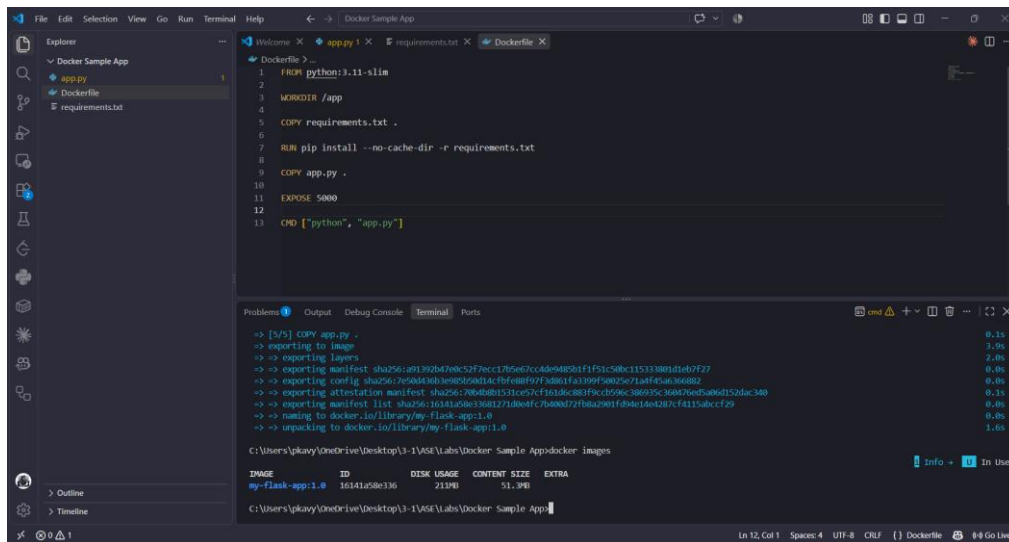
C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>docker ps

CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
4f8e2d8785d6   my-flask-app:1.0 "python app.py"         28 seconds ago Up 27 seconds 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flask-container

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>
```

7) Check the running container

- ➔ Execute command
docker ps
- ➔ Verify it's status and port mapping



8) Access the application

- ➔ Open web browser
- ➔ Enter: <http://localhost:5000>
- ➔ The flask application open in web browser
- ➔ Verify the image



9) Verify container logs

- ➔ Open the terminal
- ➔ Execute: `docker logs flask-container`
- ➔ The logs show incoming requests from browser

```
Problems 1 Output Debug Console Terminal Ports
C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
4f8e2d8785d6   my-flask-app:1.0  "python app.py"         28 seconds ago Up 27 seconds  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp  flask-container

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>docker logs flask-container
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [20/Sep/2026 05:58:08] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [20/Sep/2026 05:58:09] "GET /favicon.ico HTTP/1.1" 404 -

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>
```

10) Stop the container

➔ ->Execute the command: docker stop flask-container

```
Problems 1 Output Debug Console Terminal Ports
C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>docker logs flask-container
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [20/Sep/2026 05:58:08] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [20/Sep/2026 05:58:09] "GET /favicon.ico HTTP/1.1" 404 -

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>docker stop flask-container
flask-container

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>
```

➔ Verify the container status : docker ps

```
Problems 1 Output Debug Console Terminal Ports
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [20/Sep/2026 05:58:08] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [20/Sep/2026 05:58:09] "GET /favicon.ico HTTP/1.1" 404 -

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>docker stop flask-container
flask-container

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
4f8e2d8785d6   my-flask-app:1.0  "python app.py"         2 minutes ago Exited (137) 31 seconds ago          flask-container

C:\Users\pkavy\OneDrive\Desktop\3-1\ASE\Labs\Docker Sample App>
```

➔ Container status should be stopped